

PURBANCHAL UNIVERSITY

2025

Master of Computer Application (M.C.A.)/First Semester/Final

Time: 03:00 hrs.

Full Marks: 60 /Pass Marks: 24

MCA104/MCA114: Advanced Database Management System (New/Old)

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Group A

Answer TWO questions.

2×12=24

1(a) Diagrammatically illustrate and discuss the steps involved in processing a query. [7]

(b) Explain with example, evaluation of query with Sorting and Nested Loop. [5]

2. Explain Lock-based protocol with example. What are pitfalls of lock-based protocol? Describe Two-phase Locking Protocol. [5+2+5]

3. Consider the relational database:

employee (eid, name, address, job, salary)

works (eid, cid)

company (cid, cname, address)

Write SQL statements to:

(i) Count the number of employees who works for Creative IT Solutions Pvt. Ltd.

(ii) Display employees who are not working for Creative IT Solutions Pvt. Ltd.

(iii) Show employee name, job, salary and company name where salary greater than 35000.

(iv) Delete employees whose salary is greater than average salary of employee.

(v) Increase salary of employee by 10% who works in IT department.

(vi) Find Maximum salary of employee jobwise.

Group B

Answer SIX questions.

6×6=36

4. Explain database design for an ORDBMS.

5. Discuss various ways of accessing database from application programs. Explain about SOAP and JSON.

Contd. ...

(2)

6. ✓ What is Database Security and what are its challenges? Explain Authentication and Authorization.
7. ✓ What is Serializable schedule? Explain serializability and its types.
8. Explain Log-based recovery scheme along with deferred modification and immediate modification policies.
9. ✓ Explain the concept of RAID and B-tree index.
10. Consider the following insurance database, where primary keys are underlined:

Student (StuName, Stuld, Class, Major)

Course (CourseName, CourseNumber, CreditHours, Department)

Section (SectionId, CourseNumber, Semester, Year, Instructor-Name)

GradeReport (Stuld, SectionId, Grade)

Prerequisite (CourseNumber, PrerequisiteNumber)

Construct the following RA expressions for this relational database.

- (a) Display name and Id of those students whose major is finance and study in 4th semester.
- (b) Display the number of courses taught by 'Bhupi'.
- (c) Change credit hours of course which course number equal to 19 or 44.
- (d) Insert a new student {'Raju', 10, 12, 'DBMS'} in student relation.

11. ✓ Write short notes on:

[2×3=6]

- (i) Spatial Database and Multimedia Database
- (ii) Data Mining and Web Database

